



Research paper

Human–robot interaction-oriented video understanding of human actions

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ABSTRACT

This paper focuses on action recognition tasks oriented to the field of human–robot interaction, which is one of the major challenges in the robotic video understanding field. Previous approaches focus on designing temporal models, lack the ability to capture motion information and build contextual correlation models. This may result in robots being unable to effectively understand long-term video actions. To solve these two problems, this paper propose a novel video understanding framework including: an Adaptive Temporal Sensitivity and Motion Capture Network (ATSMC-Net) and a contextual scene reasoning module called Knowledge Function Graph Module (KFG-Module). The proposed ATSMC-Net can adaptively adjust the frame-level and pixel-level sensitive regions of temporal features to effectively capture motion information. To fuse contextual scene information for cross-temporal inference, the KFG-Module is introduced to achieve fine-grained video understanding based on the relationship between objects and actions. We evaluate the method using three public video understanding benchmarks, including Something-Something-V1&V2 and HMDB51. In addition, we present a dataset with real-world application scenarios of human–robot interactions to verify the effectiveness of our approach on mobile robots. The experimental results show that the proposed method can significantly improve the video understanding of the robots.

1. Introduction

The rapid development of computer vision has resulted in more challenges to robotic video understanding tasks in human–robot interaction-oriented domains. In the last decade, owing to the development of deep neural networks, The efficient video understanding has progressed significantly (Wang et al., 2018; Lu et al., 2022; Li et al., 2023) in areas such as robotic automatic navigation, industrial Internet of Things, electronic services and industrial assembly. For operational scenarios pertaining to human–robot interactions, the advantages of robotic systems over manuals include the ability to perform delicate operations, high loads, and rapid calculations, etc. However, current robots cannot insufficiently understanding of human behavioral activities, whereas control commands largely rely primarily fixed programs, therefore they cannot self-adapt to new operational targets and environments. Addressing the challenges above and allowing robotic systems effectively understand human actions via vision remains a significant research problem in the field of human–robot interaction.

With the booming development of deep learning techniques, purely vision-based video understanding frameworks have become increasingly sophisticated (Wu and Krahenbuhl, 2021; Wang et al., 2020; Zhang et al., 2021; Zhu et al., 2023). However, solution to this problem is hindered by the low accuracy of dynamic action recognition (Muhammad et al., 2020). Unlike static images, dynamic actions have complex

temporal correlations, therefore efficient temporal modeling is the key to solving this problem (Liu et al., 2021). Two-stream networks and 3D convolutional networks are two classical temporal modeling mechanisms. Two-stream networks can additionally supplement the action information by adding optical flow maps, which effectively improves the accuracy of action recognition. However, the acquisition of optical flow maps requires extremely high computational resources; Meanwhile, 3D convolutional neural networks can combine local temporal and spatial information for fusion, their computational efficiency is unsatisfactory, many parameters are required, and optimization is difficult to achieve. Therefore, one of the keys to solving the dynamic action recognition problem in video understanding is to design a simple and efficient temporal modeling approach, particularly for robotic devices oriented toward human–robot interaction scenarios.

In addition, logical relationships between actions and objects can be used to parse human behavioral activities in videos. For example, if a person is observed picking up a screwdriver during an assembly operation, then one is likely to assume that the screwdriver will be used for screwing, which stems from pre-learned knowledge in the human brain. This logical reasoning of a priori knowledge is the bases for the knowledge function graph model. Jelodar et al. (2018) integrate object location, object flow features and their consistency into a functional

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object-oriented knowledge representation for action recognition. Hu et al. (2015) propose an exemplar-based recognition framework for solving action recognition in human-object interaction. Dai et al. (2021) integrate object location, object flow features and their consistency into a functional object-oriented knowledge representation for video understanding. However, the problem with these approaches is that they only establish static associations for task objects and lack dynamic scenario contextual associations between actions and objects. Using the relationship between actions and objects to construct knowledge function graphs can be more effective for understanding human behaviors and instructions.

Based on the problems of the above two categories of methods, this paper aims to provide a novel video understanding framework for action recognition problems in the field of human-robot interaction. This video understanding framework can be effectively used to model video temporal information, including a lightweight Adaptive Temporal Sensitivity and Motion Capture Network (ATSMC-Net), and a Knowledge Function Graph Module (KFG-Module) for scene contextual reasoning. The ATSMC-Net can accurately recognize human behavioral actions in work scenes, and the KFG-Module can assist in task reasoning for complex scenes and in understanding human actions and commands.

Specifically, the ATSMC-Net can process the input video stream information in real time and extract temporal action features for classification. The ATSMC-Net comprises of a Motion Capture Network (MC-Net) and an Adaptive Temporal Sensitivity Network (ATS-Net). The MC-Net can effectively extract the motion information from video frames and enhance its ability to process motion features. Meanwhile, the ATS-Net enhances the expression of cross-frame features in a network and alleviates the feature loss caused by long action durations. The design of the ATSMC-Net is based on a 2D convolutional backbone, which reduces the computational effort compared with two-stream networks and 3D convolution.

Reasoning involving static objects cannot be accomplished using only on the ATSMC-Net to obtain the results of behavioral actions. The implications of an action can vary depending on the scenario. For example, merely recognizing the action “take” does not allow one to understand the specific behavior, in this case, “what was taken” may be the solution. The purpose of the KFG-Module is to solve the problem of fine-grained action understanding based on the relationships between scenes and actions. The KFG-Module comprises an Environment Aware Network (EA-Net) and a Dynamic Reasoning Graph (DR-Graph). The EA-Net is used to identify objects and obtain depth information from a scene. The DR-Graph is used to search for logical relationships that exist between objects and active objects, and then generate video understanding results based on scene contextual information.

In summary, the contributions of this study are as follows:

- (1) The proposed ATSMC-Net solves the problem where the model cannot effectively capture motion features in dynamic video sequences, thus improving the accuracy of model video understanding.
- (2) A novel KFG-Model is introduced to implement object-action video understanding in an end-to-end manner based on contextual association of scenes.
- (3) We present an end-to-end framework for robotic video understanding in the field of human-robot interaction and achieve state-of-the-art performance on multiple publicly available benchmarks and self-developed datasets. In addition, the approach is demonstrated to be effective for robot manipulation through real-world scenario testing.

The remainder of this paper is organized as follows. Section 2 describes the related video understanding methods. Our approach is presented in Section 3. Section 4 presents the experimental setup and detailed experimental results. Finally, Section 5 presents the conclusions.

2. Related work

2.1. Human action recognition

The methods of human action recognition can be roughly divided into three categories from the perspective of feature extraction: two-stream network based methods, 3D convolutional neural network based methods, and variants of classical LSTM methods. Wu et al. (2021b) proposes a novel multimodal two-stream 3D network framework that can exploit complementary multimodal information to improve recognition performance. Simonyan and Zisserman (2014) proposes a two-stream ConvNet architecture containing spatial and temporal networks for recognizing video actions. Carreira and Zisserman (2017) introduce a new Two-Stream Inflated 3D ConvNet based on 2D ConvNet inflation, which allowed the spatiotemporal features of a video to be extracted in an end-to-end manner. Industrial systems feature complex action recognition tasks, thus rendering temporal reasoning is particularly important. Wang et al. (2023a) proposes a novel meta-negativity network that uses a contrastive learning scheme to alleviate insufficient discriminative information, thus achieving accurate behavior recognition. Li et al. (2021a) proposes a resilient semantic network to learn new actions by decomposing the human body into several semantic body parts. In addition, several lightweight identification frameworks (Ullah et al., 2021) are being developed for industrial electronics applications. To further reduce the computational effort, several methods (Jiang et al., 2019; Li et al., 2020a; Sudhakaran et al., 2020) to balance accuracy and speed are proposed. Lin et al. (2019) shifts in the channel dimension to temporal modeling and obtains a performance close to that of 3D convolution without increasing the computational effort. Tran et al. (2018) decomposes the 3D convolution into 2D convolution and 1D convolution to reduce the amount of computation. In this study, we design a lightweight model that exploits the strengths of these approaches and deploy it on a mobile robotics platform.

2.2. Scene contextual inference for video understanding

The interaction between actions and objects forms the structural information of continuous actions, that can be classified into multiple active elements and used for video understanding (Jelodar et al., 2018; Ryu et al., 2021). We can reorganize active elements based on the scenario and render them applicable to different contextual reasoning tasks. Human actions are typically combinatorial in pattern, i.e., they comprise subjects, objects, and spatiotemporal interactions between them. Fan et al. (2021) construct scene interaction information by modeling the states of hands, objects, and reference objects as well as their interactions with each other. Yang et al. (2023) propose a graphical relational decision network that can mine the relationships between objects in a dataset to improve recognition accuracy. Materzynska et al. (2020) propose an object-temporal interaction network that can explicitly model the spatially varying relationships between action subjects and targets, thus solving the problem in which existing action recognition models cannot capture the combined verb and noun relationships. Xie et al. (2023) propose a novel iterative knowledge propagation and relational reconstruction network for image text matching to obtain fine-grained intra-modal associations and features. Li et al. (2021b) introduce a sub-graphs to represent and encode the discriminative patterns of each action in the videos, they use Gaussian mixing layers to decompose the graph into subgraphs of multiple scales for classification. Li et al. (2020b) regards each object proposal detected in a sample frame as a graph node, and then searches the adaptive network structure to model object interactions. Unlike the previously mentioned relational reasoning method for video understanding, we combine multiple modalities to perform spatio-temporal reasoning in an end-to-end mode.

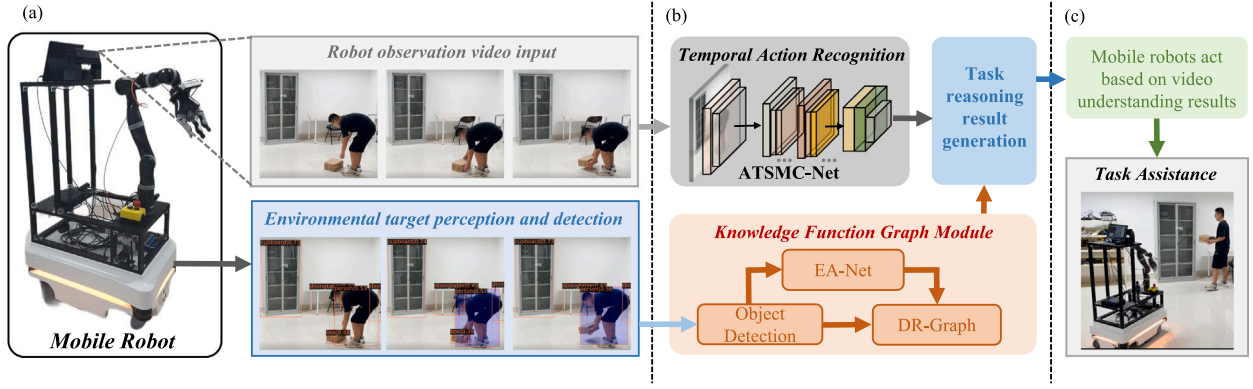


Fig. 1. The proposed framework for video understanding in mobile robots. (a) appearance of the mobile robot. Video is captured by the camera and the video input is pre-processed by the processor at the bottom. (b) Proposed methods. The ATSMC-Net performs the recognition of temporal actions and then combines with the KFG-Module, which is responsible for the contextual correlation of scenarios, to generate the final task inference results. (c) Human–robot interaction tasks. Robot completes the assistance task autonomously based on the reasoning results.

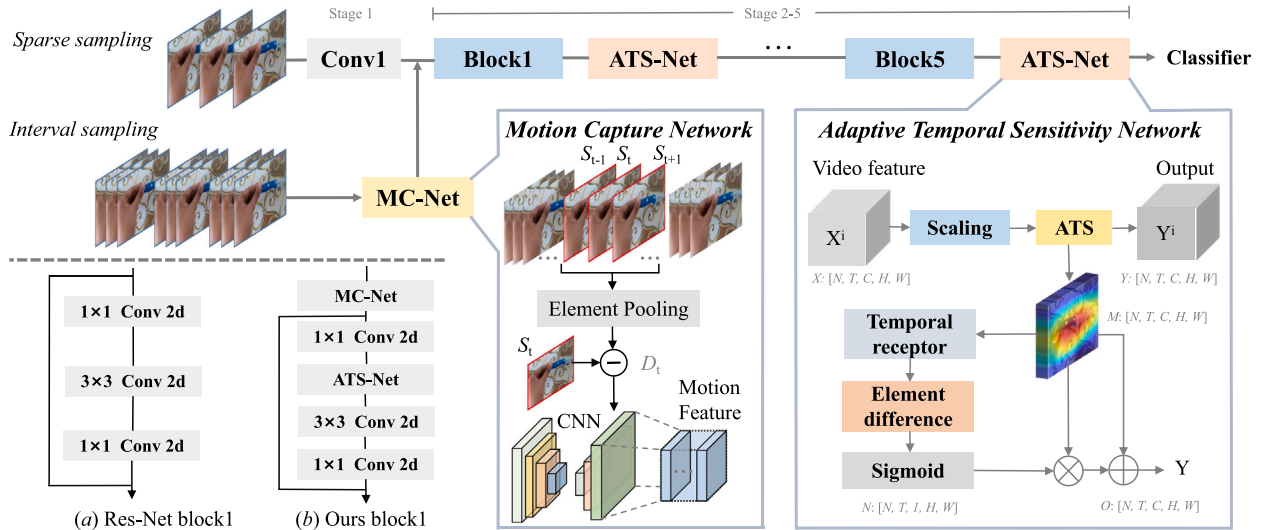


Fig. 2. Illustration of the proposed ATSMC-Net. Motion Capture Network (MC-Net) can effectively mine more motion cues in low-dimensional features and attach them to the backbone. Adaptive Temporal Sensitivity Network (ATS-Net) enhances the representation of cross-frame features in the network and alleviates the problem of feature loss due to long action durations. The MC-Net is inserted after Stage 1 of the backbone, and the ATS-Net is inserted within each module of Stages 2–5. (a): Res-Net block, (b): Our block.

3. Our method

In this paper, we aim to provide a new video understanding framework for action recognition problems in the field of human–robot interaction oriented to solve the problem of robots unable to effectively understand prolonged video actions. Therefore, we design a framework containing an ATSMC-Net and a KFG-Module to provide a basis for downstream tasks such as human–robot mutual assistance. Fig. 1 shows an overview of our robot video understanding framework.

3.1. Adaptive Temporal Sensitivity and Motion Capture Network

Video motion features contain complex temporal and spatial information, and previous methods focus excessively on spatial information to effectively model temporal information (Zhou et al., 2018). As temporal modeling research progresses, Wang et al. (2018) convert the motion cues into the form of optical flow maps for temporal modeling and achieve significant improvements in terms of classification. Kwon et al. (2021) attempt to obtain motion information between video frames, participated in temporal modeling in an additional manner, and achieve satisfactory results. This proves that, to some extent, motion cues are indispensable in temporal information modeling. Therefore, to comprehensively investigate the role of motion cues in temporal

information modeling, we propose an Adaptive Temporal Sensitivity and Motion Capture Network (ATSMC-Net).

The proposed ATSMC-Net aims to adaptively model the temporal sequences of motion cues in an input video, as shown in Fig. 2. We begin with the input of the video frame and then implement the proposed formulation as an effective block. We use the Res-Net to instantiate our network such that it can utilize the pre-trained model. Each module of the original network is illustrated in Fig. 2(a). The MC-Net serves as an additional module in Stage 1, whereas an Adaptive Temporal Sensitivity Network (ATS-Net) is embedded in each block of the backbone, as shown in Fig. 2(b).

Adaptive Temporal Sensitivity Network. Formally, the video features of Stages 2–5 are denoted as X^i with the shape $[N, T, C, H, W]$. To reduce the computation, the channels are first scaled by a 1×1 2D convolution to obtain the intermediate features M consequently the shape becomes $[N, T, C/r, H, W]$, where r indicates the scaling ratio. The temporal receptor is defined as a sensitive feature filtering unit that adaptively retains the feature values with the greatest variation. Therefore, the temporal receptor acquires the relative motion change values of the local feature frames. This process can be expressed as follows:

$$V_t = M * \sum_{i=1}^H \sum_{j=1}^W \max \{0, C_{i,j}(t|d)\}, \quad (1)$$

where, $*$ represents the convolution operation, C represents the feature value, t represents the temporal dimension, d represents the stride of the convolution and $d = 1$ is set in the experiment.

After obtaining the motion feature value V_t , it is necessary to calculate the element difference of V_t in the original video feature. It aims to eliminate the interference of static background feature values and to obtain sensitive feature values related to temporal cues. This can be expressed as follows:

$$V_{diff} = V_t - V_b, \quad (2)$$

where, V_b denotes the feature value of the current frame. V_{diff} denotes the temporally sensitive features, which represents to some extent the intensity of the motion change occurring in the current frame. Note that the features obtained here cannot be directly utilized and need to be processed using the activation function. We choose the *Sigmoid* function as the activation function here because it can suppress insensitive features by returning negative values back into the original features. The activated features are normalized to a multidimensional matrix and finally the output is obtained by multiplying the elements in an attentional manner. This process is represented as follows:

$$N = M + M \odot F'(V_{diff}), \quad (3)$$

where, $\odot$ denotes element multiplication, and F' denotes the activation function.

Motion Capture Network. The proposed ATS-Net can effectively model temporal cues from the input video clips. However, some low-dimensional motion cues are inevitably lost as the features are processed in a deep network. Although the introduction of optical-flow cues can effectively alleviate this problem, obtaining optical flow maps is difficult. Therefore, we attempt to mine the motion cues using a denser video sample. We introduce a new Motion Capture Network (MC-Net) to efficiently mine additional motion cues in low-dimensional features and attach them to the backbone.

Based on previous studies (Wang et al., 2021c; Li et al., 2020a), we first divide the dense video frames into n segments equally, and then select a frame randomly in each segment to obtain the sparsely sampled segment S^n as the input to the main network. In addition, we randomly select i consecutive frames from each segment to obtain the interval sampling segment S_i^n as the input for the MC-Net.

For the input image, changes in content between consecutive frames can be reflected by the accumulation of pixel differences (Wang et al., 2021c; Tian et al., 2022). Higher accumulation values represent a more dramatic degree of motion occurrence, and potential video feature values are more effective for motion representation.

For each sampled clip, we perform a simple element pooling operation to traverse all pixels and then create a difference with the intermediate frames to obtain the motion input D_t . Subsequently, we fed D_t into a set of convolutional neural networks for feature extraction to obtain the motion features, and finally it is added to the backbone network. The process is represented as follows:

$$F_{mc} = \text{Conv}(S_t^n - E_p[S_{t-1}^n, S_t^n, S_{t+1}^n]), \quad (4)$$

where, F_{mc} represents the features output by the MC-Net, and E_p denotes the element pooling operation.

3.2. Knowledge Function Graph Module

The scenarios of video understanding tasks involve interactions between actions and objects in addition to actions, and the video understanding tasks in complex scenarios cannot be accomplished via only action recognition. A proven approach is to segregate the video understanding task in scenes into multiple subtasks and search for the results in the form of knowledge graphs (Jelodar et al., 2018). Therefore, the KFG-Module is designed to solve the problem of fine-grained action understanding based on the relationships between scenes and action.

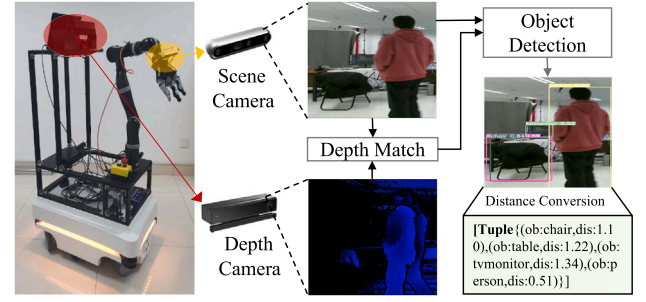


Fig. 3. The proposed environment aware network. The scene camera on the robotic arm acquires a video of the target, whereas the depth camera on top acquires the depth information of the target. The position and depth of all task targets in the environment are acquired via depth matching. The data are saved in the form of a tuple.

Environment Aware Network. EA-Net detects objects and their depth information in the environment, which is performed using a depth camera and scene camera simultaneously. The EA-Net process is illustrated in Fig. 3. First, we obtain the edges of the objects and their labels using a scene camera, and then match the regions with the depth information obtained using the depth camera. Subsequently, we calculate the relative distance s from the camera to the target object based on the scene scale. To avoid losing the depth value at the center point owing to hardware, we perform multi-point sampling around the center point, which is expressed as follows:

$$s = \gamma \sum_{i=1}^k d_i / k, \quad (5)$$

where, γ denotes the scaling factor, d_i denotes the depth value and k denotes the total number of sampling points. Finally, the object label and its relative distance information are sent to the pipeline in the form of a tuple.

Dynamic Reasoning Graph. The purpose of a dynamic reasoning graph is to correlate motion and static objects. Typically, the potential ways of correlation between targets can be set manually by human knowledge and experience in the case of simple task scenes. However, as the number of objects in the scene increases, this manual setting method will not be feasible. In this paper, we search for actions and objects by feeding them into a large linguistic knowledge base (Speer et al., 2017), and then combine them with scene information for dynamic inference, the process is shown in Fig. 4. Formally, all objects need to be associated with human nodes through action states, and the model does not perform intent inference when the action states are not active.

Despite being able to extract information regarding the location of actions and objects in a scene, the model cannot definitively determine the occurrence of interactions between objects and actions in videos. Therefore, the use of distance confidence is required to yield accurate judgments. The EA-Net provides accurate depth information for the objects in a scene, which can then be converted to the actual distance using the camera's internal reference. As shown in Fig. 5, we determine the interaction target by incremental information based on the hand-to-object distance.

This process can be expressed as follows:

$$\Delta r = r_t - r_{t-1}, 0 < t < T, \quad (6)$$

where, Δr denotes the change in distance value, T denotes the threshold value of the determination time, which usually depends on the performance of the hardware and will be described in detail in the ablation study. The current moment distance r_t is calculated as follows:

$$r_t = \sqrt{((x_t - x)^2 + (d_t - d)^2)} / d\hat{t}, \quad (7)$$

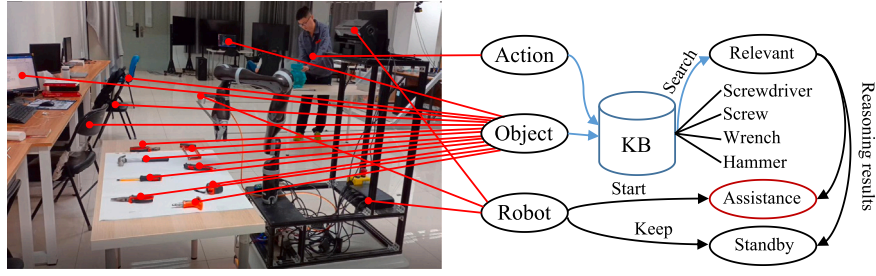


Fig. 4. Illustration of the proposed dynamic reasoning graph. KB denotes for exogenous knowledge base. Action recognition results and static targets are retrieved by KB search for all potential correlations to decide whether the robot acts or not.

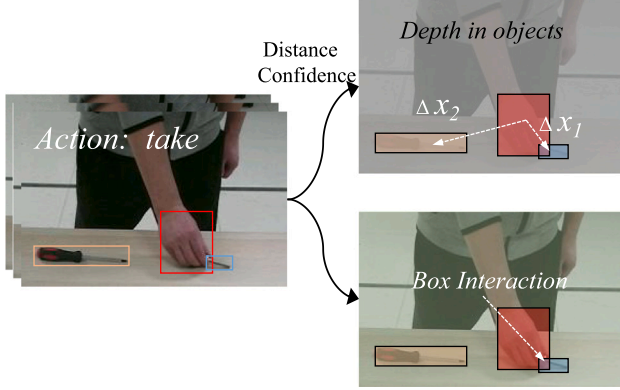


Fig. 5. The proposed method for object interaction determination. In this example, the depth information of the object is converted to a relative distance by the camera internal reference. The interaction between the human and screw is determined when an interaction occurs between the hand and the detection box of the screw.

where, x denotes the initial pixel position, d denotes the depth of the initial pixel position, and $\hat{t}$ denotes the displacement time. The hand and object are assumed to have interacted with each other when $\Delta r < 0$ is satisfied in all T .

4. Experiments and results

4.1. Datasets

To evaluate the proposed method, this study creates a new Human-robot Interaction Scene Action Recognition dataset (HISAR-dataset) to simulate action recognition during human-robot interaction. The dataset contains 14 single-person actions and 11 person object interactions with approximately 6600 videos (4640 for training and 1957 for testing), as shown in Fig. 6. Furthermore, the dataset also provides detection boxes for people and objects in a scene, which are to be used in downstream tasks. The difference with the Something dataset is that all object classes in the HISAR dataset are contained in the exogenous knowledge base, which is an important condition that is not currently available in publicly available datasets.

In addition, the proposed ATSMC-Net is tested on the public datasets Something-Something V1&V2 and HMDB-51. The Something-Something V1 dataset is a large dataset containing recorded actions between humans and some objects in daily life, with 174 categories of actions and approximately 100,000 videos (note: the V2 version contains more videos). Compared with other datasets, it focuses more on the short-term temporal modeling capability. Whereas the HMDB-51 dataset is based on internet videos clips, with a total of 6700 videos in 51 categories, HMDB-51 cannot be recognized by a static background only.

4.2. Experimental setup

Online Test. All the models are trained on 8 NVIDIA RTX 4090 with 24G bits of memory. Our model is implemented in Pytorch, and we use ResNet 50, pre-trained on ImageNet, as the backbone. For the pixel size of the sampled clips, the short sides of the frames are scaled to 256 and then cropped to 224×224 pixels. For a fair comparison, we use the hyperparameter settings of the baseline method (Wang et al., 2018). Specifically, the model uses the SGD optimizer with an initial learning rate of 0.01 for 60 epochs and decays by a factor of 10 every 10 epochs. The dropout rate is set to 0.5.

Offline Test. To verify the performance of the model on a mobile platform, we use a Kinova Gen2 7DOF robot equipped with a Kinova KG-3 gripper as a movable stand. The robot processor is equipped with an NVIDIA RTX 3090 with 24G bits of memory. Additionally, we used an Intel RealSense D435i RGB-D camera, which is mounted on the wrist of the robot and Microsoft Kinect V2 camera mounted on top of the robot. The object detection algorithm uses yolov5, and the robot arm moves and grasps using our own AFFGA-Net algorithm.

4.3. Comparison with state-of-the-arts methods

(1) *HISAR-dataset*: We first compare our method with other state-of-the-art methods on the HISAR-dataset, as shown in Table 1. The previous methods can be classified into three categories, the first is based on two-stream networks, the second is based on the 3D-CNN framework, and the third is based on the 2D-CNN framework. Our proposed method is based on the 2D-CNN framework, which facilitates the deployment of the model on mobile robots.

First, we compare several baseline methods based on two-stream networks. Wang et al. (2018) are the first to propose a two-stream model based on sparse sampling for action recognition, which achieves a classification accuracy of 88.5%. Zhou et al. (2018) improve the real-time temporal relational inference method, which enhances the classification performance of the model, and ultimately achieves a classification accuracy of 89.9%. Zolfaghari et al. (2018) propose a 3D convolutional hybrid temporal modeling method with desirable spatio-temporal representation and finally achieves 94.4% classification accuracy, however, the proposed method achieves 95.4% classification accuracy without using optical flow information, which indicates that the proposed method performs better in terms of temporal modeling. Although the performance of the original model can be improved by using the two-stream architecture, the optical flow map is difficult to obtain and is not suitable for online action recognition applications.

Next, we also compare three classical 3D-CNN models Carreira and Zisserman (2017), Zolfaghari et al. (2018) and Feichtenhofer et al. (2019). Compared to these models, our model demonstrated higher levels of classification accuracy by 4.0%, 1.7%, and 1.3%, respectively, and indicated significant advantage in terms of computational volume (153 vs. 36, 47 vs. 36, and 42 vs. 36, respectively). In addition, the 3D-CNN model is computationally intensive, which is unfavorable for deployment in mobile robots. By contrast, our approach offers advantages in terms of both accuracy and computational volume.

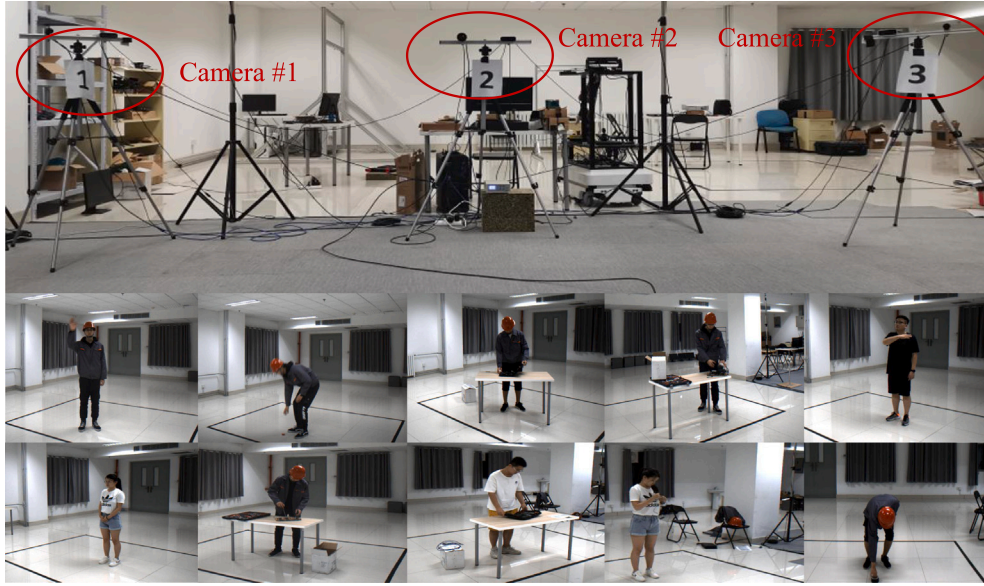


Fig. 6. Video capture devices and examples of partial datasets. Three cameras are used to capture data from three separate angles.

Table 1

Comparison Results (%) in State-of-the-art Two-stream and 2D/3D Convolution Based Methods on the HISAR dataset.

Method	Backbone	Pre-trained	GFLOPs	#Frames	Accuracy(%)
Two-stream:					
Wang et al. (2018)	BN-Inception	ImageNet	–	8+8	88.5
Zhou et al. (2018)	BN-Inception	ImageNet	–	8+8	89.9
Zolfaghari et al. (2018)	BN-Inception+3D-ResNet18	Kinetics	–	16+16	94.4
3D CNNs:					
Carreira and Zisserman (2017)	3D-ResNet50	ImageNet	153G	32	91.4
Zolfaghari et al. (2018)	BN-Inception+3D-ResNet18	Kinetics	47G	16	93.7
Feichtenhofer et al. (2019)	ResNet-50	None	42G	8	94.1
2D CNNs:					
Lin et al. (2019)	ResNet-50	Kinetics	33G	8	91.2
Jiang et al. (2019)	ResNet-50	ImageNet	34G	8	90.8
Li et al. (2020a)	ResNet-50	ImageNet	35G	8	94.2
Wang et al. (2021b)	ResNet-50	ImageNet	35G	8	92.7
Ours	ResNet-50	ImageNet	36G	8	95.4

Finally, our method outperforms other similar 2D-CNN methods in the dataset. Lin et al. (2019) uses the shift operation to obtain temporal cues and obtains a classification accuracy of 91.2%, which is 4.5% higher than that of the baseline method. However, the temporal modeling capability of this method is limited. Jiang et al. (2019), Li et al. (2020a) and Wang et al. (2021b) both construct attention enhancement modules for local features based on feature differences and achieve 90.8%, 94.2% and 92.7% classification accuracy, respectively. Jiang et al. (2019) introduces feature differences into the model in an additive way, Li et al. (2020a) improves on this by incorporating them into the model in an attentional way, and introducing 1D Conv as a complementary feature to the temporal cues. Wang et al. (2021b) additionally introduces 3D Conv as a mining module for temporal cues. However, these three methods construct feature differences in a static mode, which results in minimal differences between similarly distributed features and thus cannot be optimized further. By contrast, our method adaptively captures the regions with the greatest variation within the receptive field and incorporates these sensitive regions into the original features in the form of attention, thus maximizing the model's ability to express features in the motion region. The experimental results show that our method outperforms them by 4.6%, 1.2% and 2.7%, respectively, thus indicating that our dynamic feature difference construction method is more advantageous.

(2) *Something-Something V1&V2*: To further verify the advantages of our method in managing temporal features, we compare our method

Table 2

Performance (%) comparison on Something-Something V1&V2. Here, “–” Indicates the Numbers are not Available.

Method	FLOPs	Something V1		Something V2	
		Top-1	Top-5	Top-1	Top-5
Wang et al. (2018)	BN-Inception	19.5%	–	33.4%	–
Lin et al. (2019)	ResNet-50	45.6%	74.2%	58.8%	85.4%
Jiang et al. (2019)	ResNet-50	49.2%	79.3%	62.3%	88.8%
Li et al. (2020a)	ResNet-50	48.9%	78.1%	–	–
Liu et al. (2021)	bLResNet-50	48.4%	78.8%	61.7%	88.1%
Carreira and Zisserman (2017)	3D-ResNet50	41.6%	72.2%	–	–
Zolfaghari et al. (2018)	3D-ResNet18	46.4%	–	–	–
Wang et al. (2023b)	ResNet-50	50.4%	80.2%	62.4%	88.6%
Wu et al. (2021a)	ResNet-50	48.8%	–	60.8%	–
Shen et al. (2021)	ResNet50	48.1%	76.2%	60.1%	–
Ours	ResNet-50	50.6%	79.1%	62.9%	88.5%

with the state-of-the-art methods on Something-Something V1&V2, as shown in Table 2. Liu et al. (2021) propose an adaptive dynamic convolution kernel that extends the perceptual field of 2D convolution into 3D, achieving the top-1 classification accuracies of 48.4% and 58.8% and the top-5 classification accuracies of 78.8% and 85.4% for V1 and V2, respectively. Wang et al. (2023b) propose an adaptive multi-granularity spatio-temporal network that can effectively capture the complex-scale dynamics of spatio-temporal cues, achieving top-1

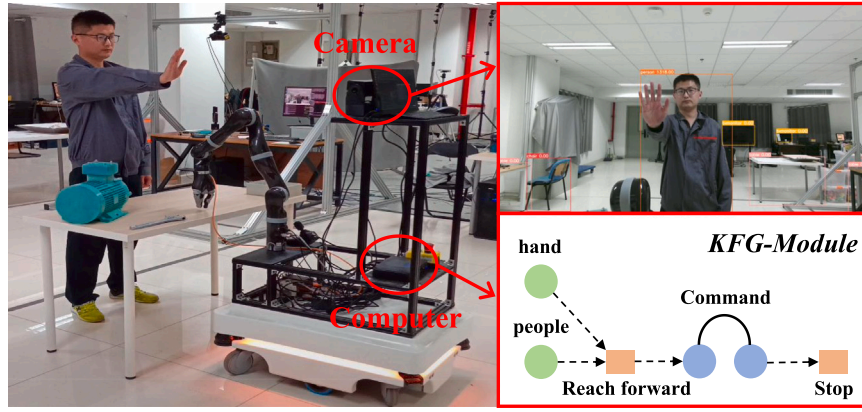


Fig. 7. Example of a real human–robot interaction scenario. The robot first observes the human action and then uses the KFG-Module to reason about the intent and execute the corresponding command.

Table 3

Comparison Results (%) in State-of-the-art on the HMDB51 Datasets.

Method	Backbone	HMDB51
Li et al. (2020a)	ResNet-50	73.3%
Carreira and Zisserman (2017)	Inception V1	74.8%
Xie et al. (2018)	Inception V1	75.9%
Zolfaghari et al. (2018)	Kinetics	72.4%
Wu et al. (2021a)	ResNet-50	75.7%
Zhang et al. (2020)	ResNet-50	72.4%
Li et al. (2021b)	ResNet-50	72.2%
Ullah et al. (2018)	MobileNet V2	72.2%
Wang et al. (2022)	ResNet-50	75.2%
Ming et al. (2023)	Frequency	73.8%
Ours	ResNet-50	76.1%

Table 4

Performance of robot video understanding in real scenarios.

Method	Mode	Accuracy
Baseline (Ullah et al., 2018)	RGB	41.2%
Baseline (Ullah et al., 2018) + KFG-Module	RGB	81.3%
ATSMC-Net + KFG-Module	RGB	93.4%
ATSMC-Net + KFG-Module	RGB+D	94.1%

classification accuracies of 50.4% and 62.4%, respectively. Wu et al. (2021a) introduce a new multi-view fusion module that uses separable convolution's to improve video dynamic feature extraction efficiency. However, this method sacrifices accuracy to some extent. The results indicate that our method outperforms other approaches significantly in terms of Top-1 classification accuracy and surpasses most methods in terms of Top-5 classification accuracy. This shows that the proposed is highly advantageous in handling temporally correlated events. Combining this with the computational efficiency advantage of the 2D-CNN framework approach, our method achieves a good trade-off between accuracy and computational efficiency.

(3) *HMDB51*: To validate the effectiveness of our method in capturing object appearance information and feature correlation, we compare our method with other state-of-the-art methods on the HMDB-51 datasets. As shown in Table 3, our method achieves the best performance on the HMDB-51 dataset. Xie et al. (2018) try to split the 3D convolution into a combination of 2D convolution and 1D convolution, but this approach sacrifices some temporal modeling performance of 3D convolutions to enhance its capability in capturing appearance features. Furthermore, this approach yields worse results than our method by 0.2% on the HMDB-51 dataset. Li et al. (2021b) propose a new subgraph design to represent and encode the distinguishing pattern of each action in the video, and the classification accuracy of this method is 72.2%, which is 3.9% lower than that of our method

owing to the absence of temporal modeling. Ullah et al. (2018) input both optical flow information and temporal information into an LSTM network for recognition tasks. However, the method (Ullah et al., 2018) has poorer results than the convolution-based method, achieving only 72.2% in terms of classification performances. Wang et al. (2022) design a spatio-temporally effective coding module and introduce a novel Twins training framework that achieves a classification accuracy of 75.2%. Ming et al. (2023) propose a frequency-based action recognition method that achieves a classification accuracy of 73.8 by enhancing low-frequency patterns and edge cues. Compared with these methods, our approach performs well on the HMDB-51 dataset. The HMDB-51 dataset contains not only static background information but also rich temporal information. Although our method focuses more on modeling temporal information and capturing motion clues, the results show that our approach can balance appearance and temporal information more effectively.

4.4. Realistic scenario testing in the field of human–robot interaction

To evaluate the video understanding performance in real-world scenarios, we simulate human–robot interaction operation scenarios in a laboratory for testing. We use cameras on mobile robots to observe human action and objects in work scenes, and then generate the target intent using our method to perform intent-related human–robot interaction tasks. Fig. 7 shows a test scenario: the robot first observes a human attempting to reach forward and then converts it into the corresponding intention “stop” by the KFG-Module. Fig. 8 shows a simple and complete example and the complete test process is shown in Supplementary Video.¹

To provide a comprehensive demonstration, we include human–robot interaction actions corresponding to intent understanding in the demonstration video. Target localization, robot path planning, and object grasping are from other studies performed by the team (Wang et al., 2021a) and are not presented herein. The final test results are shown in Table 4, it should be noted that the accuracy rates in the table indicate the percentage of correct intent understanding results produced by the model, not the accuracy rate of action recognition. The accuracy of video understanding with only the baseline method is only 41.2%, because the pure action recognition method could not perform the temporal inference task. The addition of the KFG-Module shows a significant improvement in the results, reaching 81.3%. This proves that the KFG-Module can perform long temporal inference tasks very well. The third row of the table shows the video understanding results after replacing the ATSMC-Net; as shown, a high accuracy of

¹ <https://youtu.be/FhuEoPaQAg4>

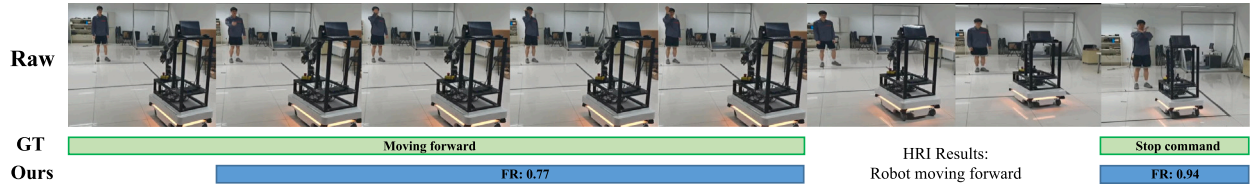


Fig. 8. Example of human-computer interaction. This example shows the robot moving autonomously after recognizing a forward motion and finally terminating after recognizing a stop motion. GT represents the ground truth, FR represents the percentage of correctly recognized frames to all ground truth.

Table 5

Detail structural ablation study of Adaptive Temporal Sensitivity and Motion Capture Network.

Method	GFLOPs	Accuracy
Baseline (Ullah et al., 2018)	16.1	86.7%
+ MC-Net	17.3	92.7%
+ ATS-Net	35.2	94.2%
ATSMC-Net	36.4	95.4%

Table 6

Performance of ablation studies with Adaptive Temporal Sensitivity Network in different phases.

Stages	Blocks	Top-1	GFLOPs
w/o ATS-Net	0	86.7%	16.1G
Res{1}	3	92.7%	19.7G
Res{1,2}	7	93.5%	24.5G
Res{1,2,3}	13	94.0%	31.7G
All	16	94.2%	35.2G

93.4% is obtained, as our method can capture motion information more efficiently and improve the overall recognition accuracy as compared with the other methods. In addition, the accuracy can reach 94.1% after combining depth information. The depth information can calculate the relative distance of the model more accurately and improve the calculation accuracy of the distance confidence. The experimental results above demonstrate the excellent adaptability of our proposed method in practical tests.

Limitations discussion. Although the task can be successfully completed in real tests, the method still possesses some limitations. First, KFG-Module is limited by the content of the exogenous knowledge base and cannot interact with some objects in the scene, which makes it difficult to directly apply the current publicly available dataset to the human-computer interaction task. If this problem is to be solved, the best way to avoid the lack of knowledge is to choose a more powerful exogenous knowledge base. Accordingly, this shows the importance of constructing HISAR-like datasets, and constructing corresponding scenarios according to the actual needs can also solve this problem effectively. Secondly, the method is a novel end-to-end action recognition approach and thus has high hardware requirements for the device, which limits the application in a wide range of mobile vision robots. Finally, unlike methods such as something-else (Materzynska et al., 2020), the detection of objects is performed through a separate network, and thus the response time of the method can be slow in mobile devices. Additionally, the complexity of the external environment can limit the KFG module's ability to interact with certain targets, and the use of more advanced object detection algorithms (Yang et al., 2023) and high-performance deep cameras would be more helpful. In summary, future work will focus on addressing these weaknesses.

4.5. Ablation study

In this subsection, we perform an ablation study to evaluate the impact of each component on the performance.

Hyperparameter Optimization. We test the optimization performance of hyperparameters, including learning rate, learning strategy

Table 7

Performance comparison of different learning rates and learning strategies.

Strategy	Initial learning rate		
	1e-2	1e-3	1e-4
MultiStepLR	94.2%	94.0%	92.3%
LinearLR	93.4%	93.2%	88.8%
StepLR	94.0%	93.9%	92.1%
CosineLR	93.7%	94.1%	92.6%

dropout and batch-size. The test results are shown in the Table 7. The performance of the different learning strategies is relatively close at learning rates of 1e-2 and 1e-3, and the performance of the different learning strategies decreases substantially when the learning rate is reduced to 1e-4. From the comparison results, the performance of 1e-2 learning rate is more stable and excellent. Dropout and batch-size are shown in Fig. 9, the best performance is achieved when the dropout is at 0.5, and increasing or decreasing it results in loss of accuracy. The best performance is achieved when batch-size is 64 with constant learning rate, and the performance decreases when increasing or decreasing, which requires adjustments to the learning rate based on the linear principle.

Impact of the ATSMC-Net. To analyze the significance of the ATSMC-Net, ablation experiments are conducted on the produced dataset. The results are summarized in Table 5. The recognition accuracy of the model after introducing the MC-Net increased by 6% compared to the baseline, reaching 92.7%, while the computation cost increase by 1.2 GFLOPs. The MC-Net utilizes the motion changes of local frames to mine potentially effective features and feeds them back to the backbone network to provide motion information, and the improvement in classification accuracy proves its effectiveness. The recognition accuracy of the model after introducing ATS-Net increase by 7.5%, reaching 94.2%, whereas the computational cost increased by 19.1 GFLOPs. The ATS-Net enhances the adaptive representation of features in the network by capturing feature differences across frames, mitigating the problem of feature loss due to the long duration of the motion, thus improving the classification accuracy. Finally, the simultaneous use of the MC-Net and ATS-Net improved the performance by 95.4% as compared with the baseline, which proves the complementarity of the two modules.

Strategy for ATS-Net stage. We also perform ablation studies on ATS-Net at different stages. The results are listed in Table 6. The recognition accuracy without the ATS-Net model is 86.7%. As the number of the ATS-Net modules inserted into the model increases, the accuracy gradually improves. At the same time, the amount of computation is gradually increasing, because the temporal feature information extracted by ATS-Net increases accordingly. When all the ATS-Net modules are fully incorporated into the backbone, the highest accuracy of 94.2% is achieved, and the computational volume reaches 35.2 GFLOPs. This indicates that the ATS-Net is effective in improving the performance of the model.

Time Threshold. To verify the effect of time thresholds on recognition accuracy, we perform tests on the produced dataset. The value of the decision time threshold T is affected by the camera and other hardware devices, so it is necessary to use empirical methods to try

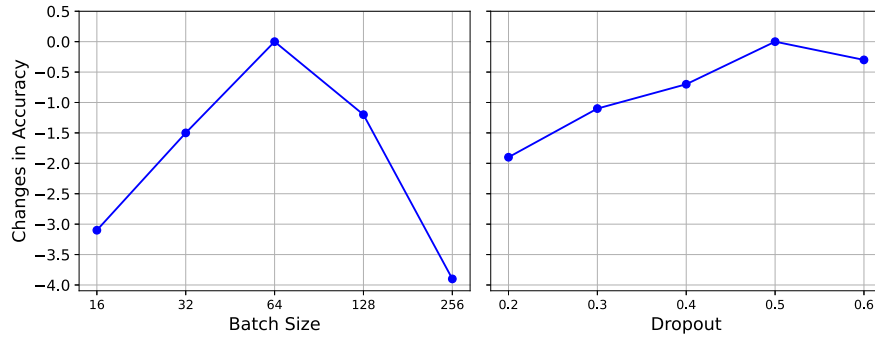


Fig. 9. Impact of Dropout and Batch-size on Performance. The default learning rate is $1e-2$.

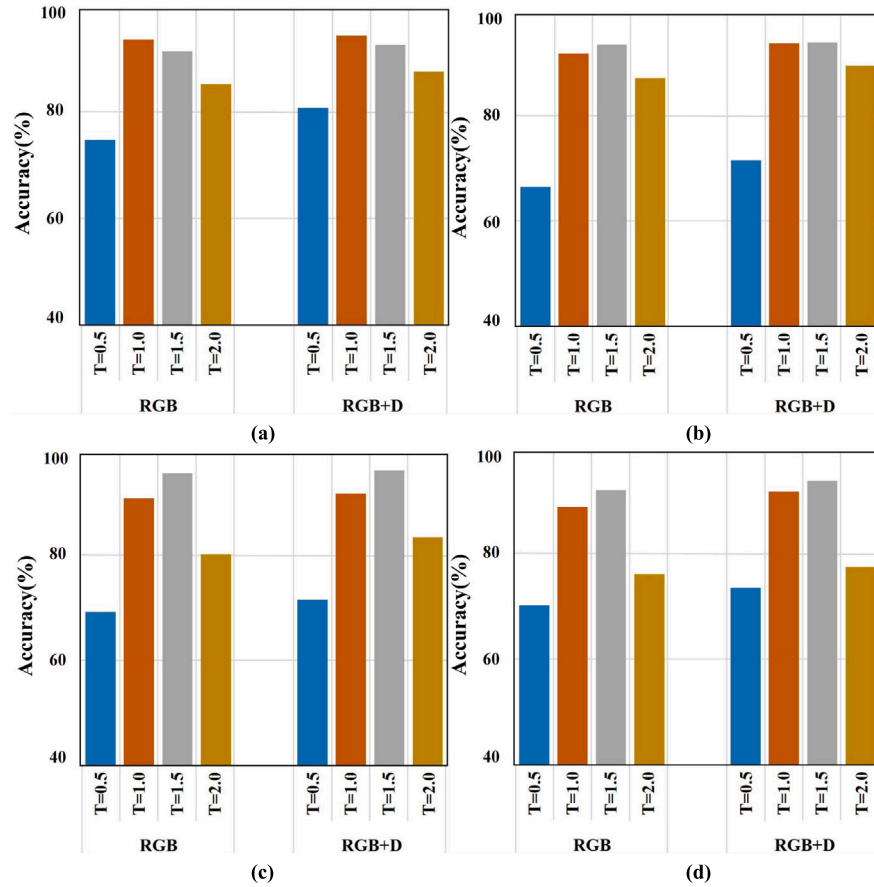


Fig. 10. The effect of time threshold selection on recognition accuracy. (a) represents video duration in 0.0–5.0 s, (b) represents video duration in 5.0–10.0 s, (c) represents video duration over 10 s, (d) represents all datasets.

and find the most suitable value for the experimental environment. We divide the dataset into three sections based on the video duration, and the results are shown in Fig. 10. The first three figures represent short time (0.0–5.0 s), medium time (5.0–10.0 s) and long time (>10 s), respectively, and the last figure is the average recognition accuracy of the entire dataset. As shown the effect is worst when the time threshold is set as 0.5, whereas it is better when the threshold is 1 or 1.5, and the effect also decreases when the time threshold reaches 2.0. This is because the number of video frames captured when the time threshold is too small is not enough for the model to fully utilize its performance. When the time threshold is extremely large, critical actions may be overlooked. In addition, the performance of the additional depth information is better than that by pure RGB because the depth information can improve the accuracy of distance confidence calculation.

Different Architecture.

We test the performance of the model under different backbone networks and the results are shown in Table 8. All tests are without pre-training. Among them, Res2Net achieves the best classification accuracy of 92.3%. SE-Net has the worst results because the attention mechanism of SE-Net conflicts with ATS-Net. In the ResNet series, the classification accuracy gradually increased as the number of network layers deepened. The highest classification accuracy of 93.4% is achieved in ResNet-101. Usually the deepening of the number of layers will additionally increase the corresponding computation amount, and the corresponding backbone network can be selected according to the actual situation.

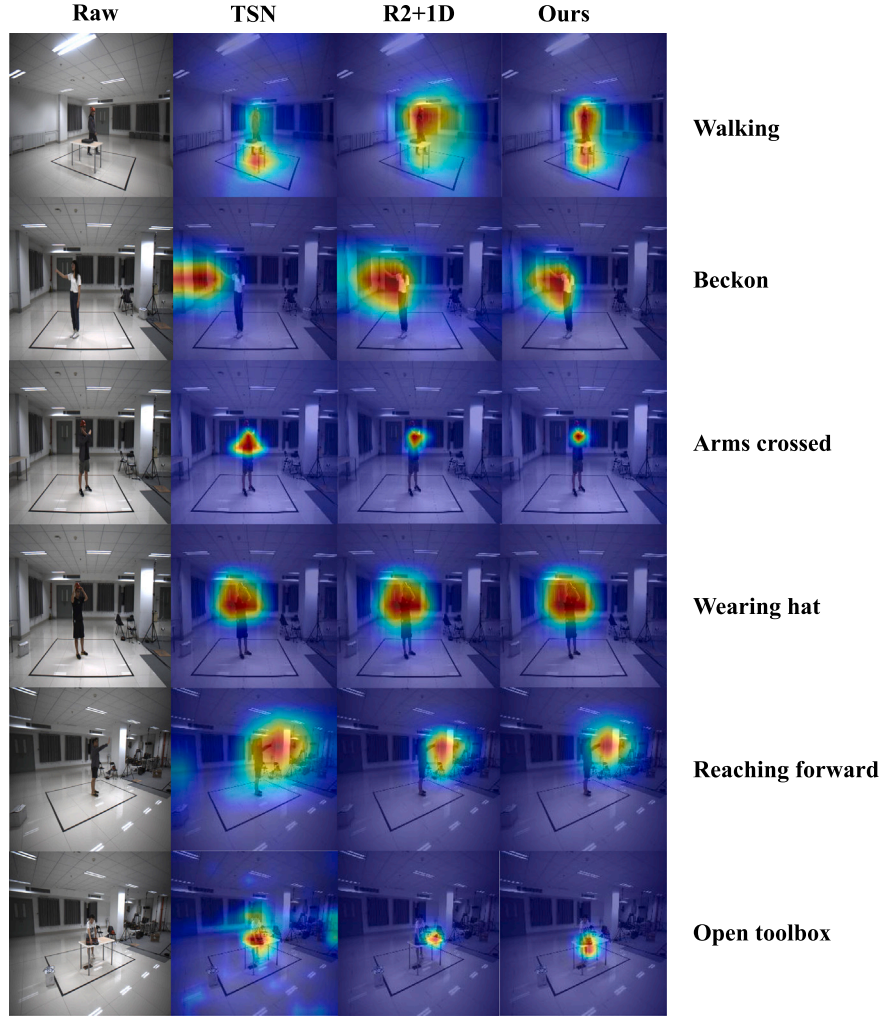


Fig. 11. Visualization of the activation maps for all predictions based on the last convolutional layer of Conv3 using Grad-CAM++. The first column of each activity shows the raw frames imported into the network. In the second and third columns, we plot the activation maps for TSN and R2+1D, respectively. And in the last column we plot the activation maps of our method.

Table 8

Performance of ablation studies with Different CNN Architecture. All Methods Without Pre-trained Models.

Stages	Frames $\times$ Crops $\times$ Clips	Top-1
BNInception	$8 \times 1 \times 1$	88.8%
SE-Net	$8 \times 1 \times 1$	87.5%
bLResNet	$8 \times 1 \times 1$	90.2%
Res2Net	$8 \times 1 \times 1$	92.3%
TSM _{Shift}	$8 \times 1 \times 1$	91.7%
ResNet-18	$8 \times 1 \times 1$	83.4%
ResNet-34	$8 \times 1 \times 1$	85.7%
ResNet-50	$8 \times 1 \times 1$	92.1%
ResNet-101	$8 \times 1 \times 1$	93.4%

4.6. Visualization

This section provides the visualization results of the class-based saliency mapping in the hidden layer of the model trained on the proposed dataset to explain the learned spatio-temporal motion features.

We visualize the class activation mapping using Grad-CAM++ (Tran et al., 2018), and the results are shown in Fig. 11. The red and blue areas correspond to stronger and weaker class correlations, respectively. We chose TSN and R2+1D as benchmarks for comparison. The

salient regions are highlighted in the frames based on the color of the pixels. TSN is more sensitive to spatial information and less sensitive to motion regions, and the heat map contains a significant amount of noise. Compared with TSN, R2+1D has a significant improvement in the motion region and can capture the motion features of video frames well. Our method achieves the best performance compared with the other methods; furthermore, the attention of the model can be focused more precisely on the motion regions with the least noise. Overall, it can be observed that for all types of motion, the motion features learned by our model are more semantically relevant and fit our cognitive model more closely than the other models.

5. Conclusion

In this paper, we propose an action recognition network named the Adaptive Temporal Sensitivity and Motion Capture Network (ATSMC-Net) and a contextual scene inference module called Knowledge Function Graph Module (KFG-Module), respectively. Specifically, the proposed ATSMC-Net can adaptively adjust the frame-level and pixel-level sensitive regions of temporal features to effectively capture motion information. This effectively improves the ability of the model in recognizing human behavioral motions to support downstream tasks. In addition, to fuse scene contextual information for cross-temporal

inference, we introduce the KFG-Module to achieve fine-grained video understanding based on the relationship between scenes and actions. This overcomes the difficulty of reasoning for multiple target interactions in video understanding tasks. The experimental results show that the method achieves state-of-the-art video understanding accuracy on several datasets and achieved outstanding recognition performance in actual experimental scenarios.

Our future endeavors are as follows: First, we plan to identify better video spatio-temporal feature capture methods to achieve lighter and more precise video understanding with a better balance of accuracy and efficiency. Second, the knowledge inference method in this paper still suffers from lack of knowledge sources, blurred target-action demarcation, etc. Hence, we shall investigate more comprehensive scene-context inference methods in the future. Finally, we will set up more experimental environments and test the algorithms in more robot models.

CRedit authorship contribution statement

Bin Wang: Writing – original draft, Writing – review & editing. **Faliang Chang:** Funding acquisition, Project administration. **Chunsheng Liu:** Supervision. **Wenqian Wang:** Investigation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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